1. ***This is the start of a multipart question. The information below pertains to question 1 to question 4.***

   WMT is the ticker symbol for Walmart. Consider the monthly returns for this stock between January 01, 2013 and May 31, 2026.

   The following code creates a vector `x` containing the monthly returns:

```
library(quantmod)
library(dplyr)

x = getSymbols("WMT",
               from = "2013-01-01",
               to = "2026-05-31",
               auto.assign = FALSE) %>%
  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  pull(monthly.returns)
```

   The returns are calculated using the adjusted closing price, which accounts for dividends and stock splits. `x` should contain 161 monthly returns.

   What was the maximum monthly return over this time period?

   a. 17.19%
   b. 14.78%
   c. 23.43%
   d. 7.30%
   e. None of the above

2. Given the information above.

   What was the interquartile range of monthly returns over this time period?

   a. 3.14%
   b. 3.70%
   c. 1.49%
   d. 6.62%
   e. None of the above

3. Given the information above.

   What is the 0.9-quantile of the monthly returns?

   a. 0.30%
   b. 9.31%
   c. 10.05%
   d. 8.03%
   e. None of the above

4. Given the information above.

   What is the empirical probability that the monthly return is greater than zero?

   a. 0.6975
   b. 0.6699
   c. 0.5901
   d. 0.7787
   e. None of the above

5. ***This is the start of a multipart question. The information below pertains to question 5 to question 6.***

   MCD is the ticker symbol for McDonald's. Consider the daily returns for this stock between September 13, 2014 and May 21, 2021.

   The following code creates a vector `x` containing the daily returns:

```
library(quantmod)
library(dplyr)

x = getSymbols("MCD",
               from = "2014-09-13",
               to = "2021-05-21",
               auto.assign = FALSE) %>%
  Ad() %>%
  dailyReturn() %>%
  as.data.frame() %>%
  pull(daily.returns)
```

   The returns are calculated using the adjusted closing price, which accounts for dividends and stock splits. `x` should contain 1683 daily returns.

   We would like to compare the empirical distribution of daily returns to a normal approximation with mean 0.001 and standard deviation 0.014.

   Focus on the 0.93-quantile. This is the return value such that approximately 93% of daily returns are less than or equal to it. Which pair correctly reports the empirical 0.93-quantile and the 0.93-quantile from the normal approximation?

   a. The empirical 0.93-quantile is 1.75%, and the normal approximation 0.93-quantile is 3.08%
   b. The empirical 0.93-quantile is 1.84%, and the normal approximation 0.93-quantile is 3.77%
   c. The empirical 0.93-quantile is 1.64%, and the normal approximation 0.93-quantile is 3.08%
   d. The empirical 0.93-quantile is 1.51%, and the normal approximation 0.93-quantile is 2.17%
   e. None of the above

6. Given the information above.

   The CDF is defined as $F(a) = \Pr(x \leq a)$. In this context, $F(a)$ is the probability that the daily return is less than or equal to a. Evaluate the CDF at $a = 0.09$. Which pair correctly reports the empirical probability and the probability implied by the normal approximation?

   a. The empirical probability is 0.9988, and the normal approximation probability is 1.6029
   b. The empirical probability is 0.9988, and the normal approximation probability is 1.0000
   c. The empirical probability is 0.5128, and the normal approximation probability is 1.5782
   d. The empirical probability is 0.4701, and the normal approximation probability is 1.5782
   e. None of the above